

Recap.

- ① $P(X > s+t | X > s) = P(X > t)$
- ② Geometric distribution: $f_X(i) = p(1-p)^{i-1}$
- ③ Exponential distribution: $X \sim \text{Exp}(\lambda)$, $f_X(x) = \begin{cases} \lambda e^{-\lambda x} & x \geq 0 \\ 0 & x < 0 \end{cases}$
(λ : average speed, x : time until event happens)
- ④ Joint pdf: $P_{X,Y}(x,y) = P_Y(y) \cdot P_{X|Y}(x|y)$
Marginal pdf: $P_Y(y) = \begin{cases} \sum_x P_{X,Y}(x,y) & (\text{discrete}) \\ \int P_{X,Y}(x,y) dx & (\text{continuous}) \end{cases}$

Exercises.

Ex 1. Hint: To verify the validity of a pdf, think of the criterion of probability measure.

(i) $\int_{-\infty}^{\infty} P_R(r) dr = \int_{-3}^0 \frac{2}{15}(r+3) dr + \int_0^2 \frac{1}{5}(2-r) dr = \frac{3}{5} + \frac{2}{5} = 1$, valid because $P(\mathbb{R}) = 1$.

(ii) $F_R(r) = \int_{-\infty}^r P_R(r) dr$.

$r < -3$: $F_R(r) = 0$.

$-3 \leq r < 0$: $F_R(r) = \int_{-3}^r \frac{2}{15}(t+3) dt = \frac{2}{15} \left[\frac{t^2}{2} + 3t \right]_{-3}^r = \frac{(r+3)^2}{15}$.

$0 \leq r < 2$: $F_R(r) = F_R(0) + \int_0^r \frac{1}{5}(2-t) dt = \frac{6+4r-r^2}{10}$.

$r \geq 2$: $F_R(r) = F_R(2) = 1$.

$$F_R(r) = \begin{cases} 0 & r < -3 \\ \frac{(r+3)^2}{15} & -3 \leq r < 0 \\ \frac{6+4r-r^2}{10} & 0 \leq r < 2 \\ 1 & r \geq 2 \end{cases}$$

(iii) $r_\alpha = F_R^{-1}(\alpha)$, $F_R(r_\alpha) = P(R \leq r_\alpha) = \alpha = P(R \leq -L)$, $r_\alpha = -L$, $\forall \alpha$, $r_\alpha = L = -r_\alpha = -F_R^{-1}(\alpha)$.

(iv) To find r such that $F_R(r) = 0.05$, since $0.05 < 0.6 = F_R(0)$, $r \in [-3, 0]$.

$\frac{(r+3)^2}{15} = 0.05$, $r = -3 + \frac{\sqrt{0.75}}{2}$, $L = -r \approx 2.134$.

(v) To find r such that $F_R(r) = 0.5$, $\frac{(r+3)^2}{15} = 0.5$, $r \approx -0.2614$.

Ex 2. Hint: $\ln'(f(x)) = \frac{f'(x)}{f(x)}$.

" $\Rightarrow$ ": if $X \sim \text{Exp}(\lambda)$, $P(X > s) = 1 - F_X(s) = e^{-\lambda s}$

$P(X > t+s | X > s) = \frac{P(X > t+s \cap X > s)}{P(X > s)} = \frac{P(X > t+s)}{P(X > s)} = \frac{e^{-\lambda(t+s)}}{e^{-\lambda s}} = e^{-\lambda t} = P(X > t)$.

" $\Leftarrow$ ": Define $f(x) = P(X > x) = \int_x^\infty P_X(a) da$,

and $\varphi(s) = \frac{f'(s)}{f(s)} = \frac{P_X(s)}{P(X > s)} = P(X > t+s | X > s) = P(X > t) = \frac{P(X > t)}{P(X > 0)} = \frac{f(t)}{f(0)} = \varphi(0)$, constant.

$\varphi'(0) = 0 = \left(\frac{f'(s)}{f(s)} \right)' = \frac{f''(s) \cdot f(s) - f'(s) \cdot f'(s)}{(f(s))^2}$, $f'(t+s) \cdot f(s) = f'(s) \cdot f(t+s)$, $\frac{f'(t+s)}{f(t+s)} = \frac{f'(s)}{f(s)} = c \in \mathbb{R}$.

$\int_0^t \frac{f'(s)}{f(s)} ds = \ln(f(t)) = \int_0^t c ds = c \cdot t - c \cdot 0 = c \cdot t$. $f(t) = e^{ct}$.

$\lim_{t \rightarrow \infty} f(t) = \lim_{t \rightarrow \infty} P(X > t) = 0$, $c < 0$, $c = -\lambda$. $F_X(x) = 1 - P(X > t) = 1 - e^{-\lambda t}$, $X \sim \text{Exp}(\lambda)$.

Ex 3.

$$(i) P_Y(y) = \int_0^1 \left(\frac{x^2}{4} + \frac{y^2}{4} + \frac{xy}{6} \right) dx = \frac{3y^2 + y + 1}{12} \quad (0 \leq y \leq 2)$$

$$P_{X|Y}(x|y) = \begin{cases} \frac{P_{X,Y}(x,y)}{P_Y(y)} = \frac{3x^2 + 3y^2 + 2xy}{3y^2 + y + 1} & 0 \leq x \leq 1 \\ 0 & \text{otherwise} \end{cases}$$

$$(ii) P(X < \frac{1}{2} | Y=y) = \int_0^{\frac{1}{2}} \frac{3x^2 + 3y^2 + 2xy}{3y^2 + y + 1} dx = \frac{1}{3y^2 + y + 1} [3x^3 + 3y^2x + 2xyx]_0^{\frac{1}{2}} = \frac{12y^2 + 2y + 1}{24y^2 + 8y + 8}.$$